Identity Exercises

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1 Level1

Part1.Determine whether each equation is an identity. Answer true or false.

1. You can use the cross method.
2. Text your answer by true(if yes), false(if not)

1. $(x+2)^2 = x^2 + 4x + 4$
2. $(x-3)^2 = x^2 - 9$
3. $(a+b)(a-b) = a^2 - b^2$
4. $(x+1)(x+2) = x^2 + 3x$
5. $10r^2 + 8 = 5(2r^2 - \frac{1}{5}) - 7$
6. $x^3 - 8 = (x-2)(x^2 + 2x + 4)$
7. $(2x-5)^2 = 4x^2 - 20x + 25$
8. $x^2 - 4x = x(x-4) + 1$
9. $(x+y)^3 = x^3 + y^3$
10. $(a+1)(a-6) = a^2 - 5a - 6$

Part2.Find the unknown constans A and B in each of following identities.

Text your answer by A=...,B=...

1. $3x+5 \equiv A(x-1)+B$
2. $2x-7 \equiv A(x+3)+B$
3. $4x+1 \equiv A(2x-3)+B$
4. $6-5x \equiv A(1-x)+B$
5. $x^2+4x-2 \equiv (x-1)(x+A)+B$
6. $2x^2-3x+7 \equiv (x+2)(2x+A)+B$
7. $3x^2-5x+10 \equiv (Ax+1)(x-2)+B$
8. $2x^2+5x-6 \equiv (2x-A)(x+3)+Bx$

Part 3. Expand the following expressions.

1. $(x+4)(x-7)$
2. $(2x-3)(x+5)$
3. $(3x-2)(4x-1)$
4. $(5-2x)(3+4x)$
5. $(x+6)^2$
6. $(2x-5)^2$
7. $(4x+3)^2$
8. $(3x-1)(3x+1)$
9. $(6x+2)(6x-2)$
10. $(x^2-3)(2x+4)$
11. $(2x^2-3x+4)(x-5)$
12. $(3x-2)^2-(x+4)(2x-1)$

Part 4. Without using a calculator, use identities to find the values of the following expressions.

1. 98^2
2. 103^2
3. $57^2 - 43^2$
4. 204×196
5. $(49.5)^2$
6. $101^2 - 2 \times 101 + 1$
7. $998^2 + 4 \times 998 + 4$
8. $\dfrac{2026^2 - 2024^2}{4}$

2 Level2

Part1.Determine whether each equation is an identity. Answer true or false.

1. You can use the cross method.
2. Text your answer by true(if yes), false(if not)

1. $(2x-3)^2 = 4x^2 - 12x + 9$
2. $(3x+4)^2 = 9x^2 + 16$
3. $(2a+5b)(2a-5b) = 4a^2 - 25b^2$
4. $(2x-3)(x+4) = 2x^2 + 5x$
5. $6k^2 - 4 = 3\left(2k^2 + \dfrac{2}{3}\right) - 6$
6. $x^3 + 27 = (x+3)(x^2 - 3x + 9)$
7. $(5x+2)^2 = 25x^2 + 20x + 4$
8. $3x^2 - 6x = 3x(x-2) - 5$
9. $(a-b)^3 = a^3 - b^3$
10. $(2p-3)(p+5) = 2p^2 + 7p - 15$

Part2.Find the unknown constants A and B in each of following identities.

1. $5x - 9 \equiv A(2x + 4) + B$
2. $3x + 11 \equiv A(x - 5) + B$
3. $6x - 4 \equiv A(3x + 2) + B$
4. $8 - 7x \equiv A(2 - x) + B$
5. $x^2 - 6x + 5 \equiv (x + 2)(x + A) + B$
6. $3x^2 + 4x - 8 \equiv (x - 1)(3x + A) + B$
7. $4x^2 - 7x + 15 \equiv (Ax - 2)(x + 3) + B$
8. $3x^2 - 2x + 4 \equiv (3x - A)(x - 2) + Bx$

Part 3. Expand the following expressions.

1. $(2x - 5)(3x + 4)$
2. $(4x + 1)(2x - 6)$
3. $(5x - 3)(2x - 7)$
4. $(4 - 3x)(5 + 2x)$
5. $(3x - 4)^2$
6. $(5x + 2)^2$
7. $(2x - 7)^2$
8. $(4x + 5)(4x - 5)$
9. $(3x - 8)(3x + 8)$
10. $(x^2 + 2x - 5)(3x - 1)$
11. $(3x^2 - 2x + 5)(2x - 3)$
12. $(4x + 1)^2 - (2x - 3)(5x + 2)$

Part 4. Without using a calculator, use identities to find the values of the following expressions.

1. 197^2
2. 206^2
3. $68^2 - 32^2$
4. 307×293
5. $(60.5)^2$
6. $203^2 - 2 \times 203 \times 3 + 3^2$
7. $1997^2 + 6 \times 1997 + 9$
8. $\dfrac{3042^2 - 3038^2}{8}$

Part 5. Word problems

1. The length of a rectangle is $(3x - 4)$ cm and the width is $(2x + 5)$ cm. Expand and simplify the expression for its area.
2. A square has side $(4a - 3)$ cm. If each side is decreased by 2 cm, find the expanded form of the new area.
3. Given $m + n = 15$ and $mn = 44$. Use algebraic identity to evaluate $m^2 + n^2$.
4. Two positive integers differ by 7, their product is 30. Let the smaller number be x. Set up an equation and expand it.
5. The area of a rectangle is $6x^2 - 7x - 20$. One side is $(2x - 5)$. Find the expression of the other side.
6. Show that the difference between the squares of two consecutive odd numbers is divisible by 8. Let the two numbers be $2k - 1$ and $2k + 1$.